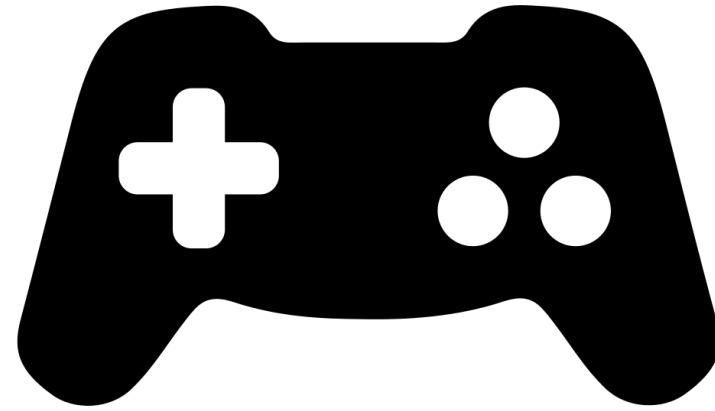


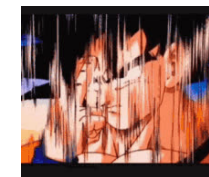


FACTRIS

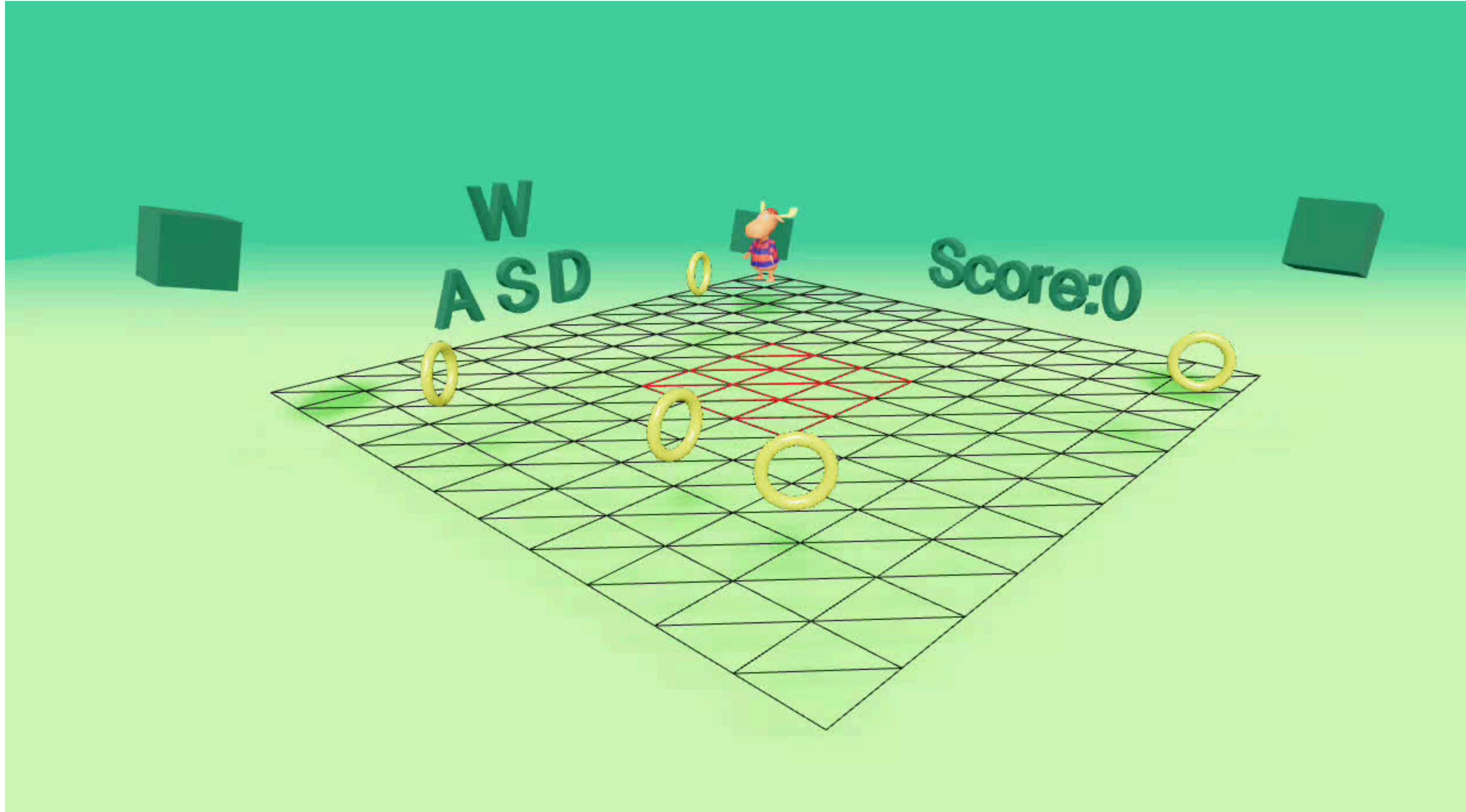
Bruno Francesco Nocera
1863075

GAME RULES

- Choice your side: **X** or **O**?
- Taking rings spawned on the map increase the score of 100 points, but, each ring can activate 1 one those special effects:
 - **Teleport effect** -> Spawn you in another cell
 - **Skip effect** -> 3 moves in which move of 2 cell instead of 1
 - **Change side** -> from X to O (or viceversa)
- When you are on a “Tris cell” you can press “Space” and then drop an X (or O).
- The goal is win the tris game while increasing the score.

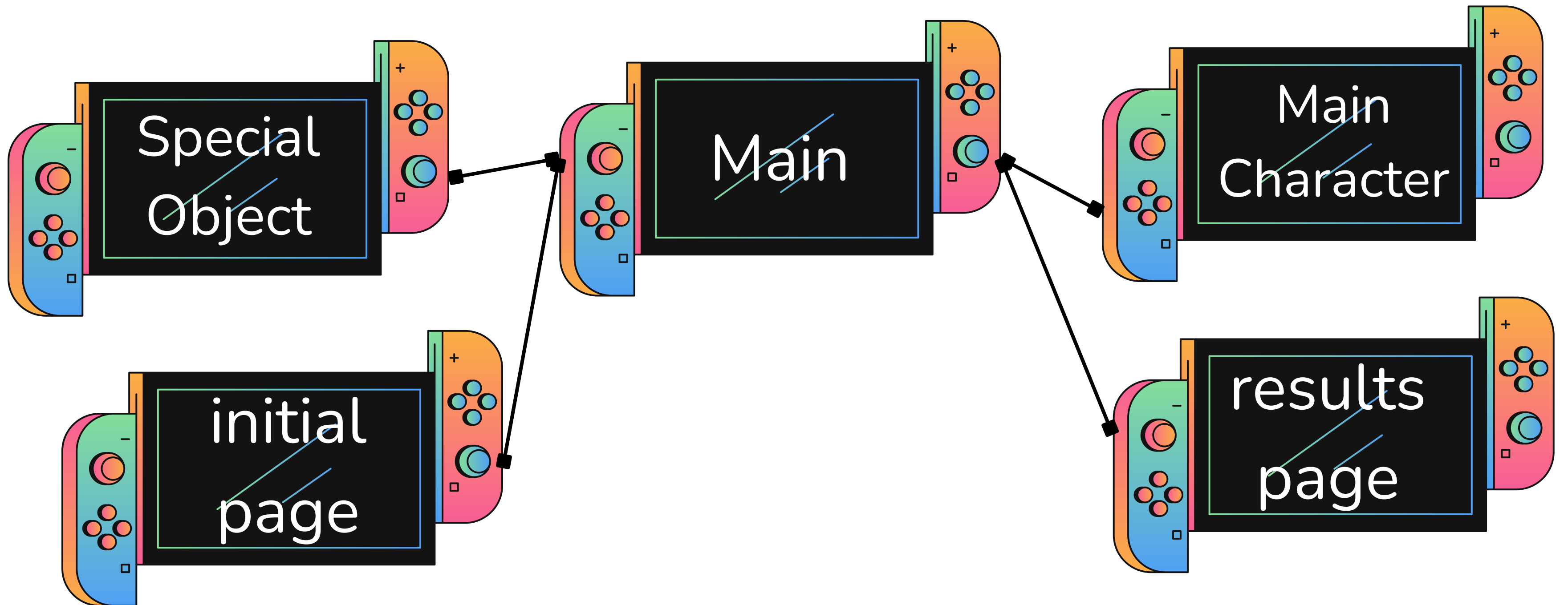


SCENE



STRUCTURE

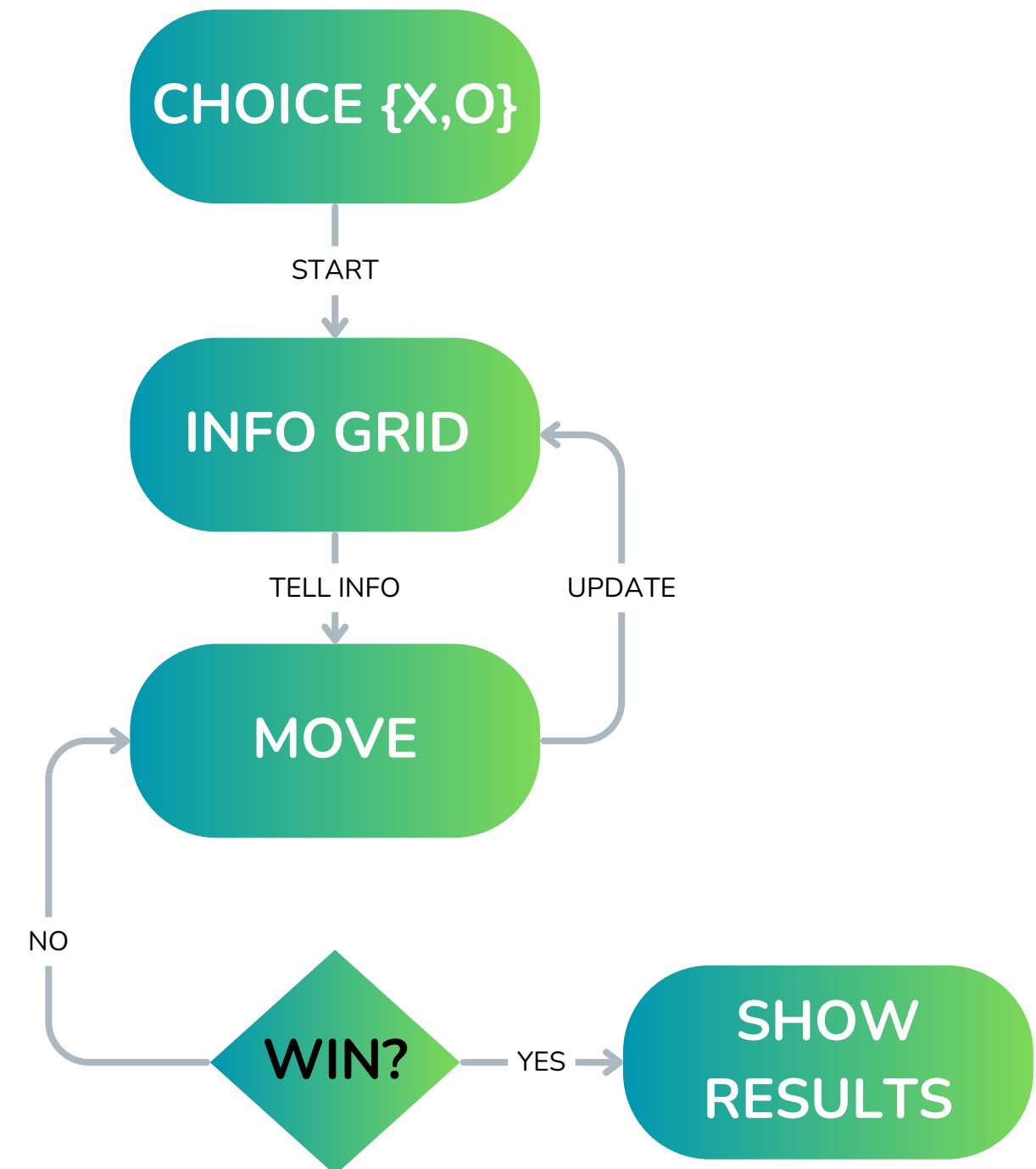
- There are 3 principal components:



GAME LOGIC

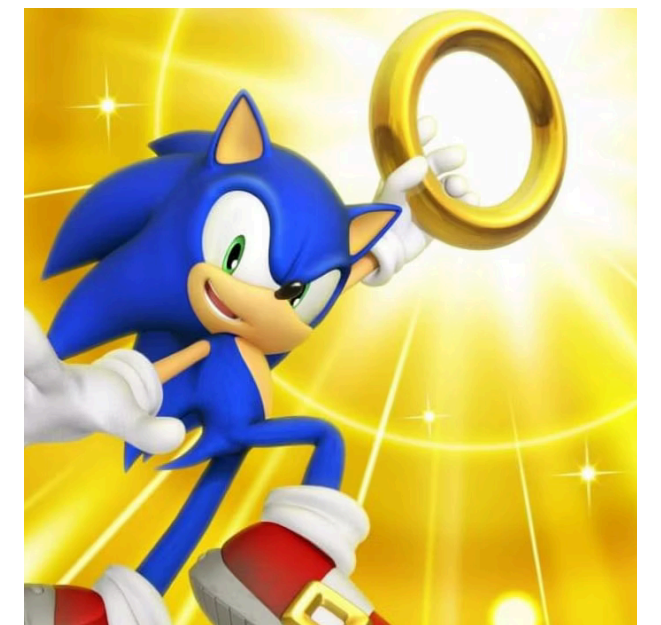
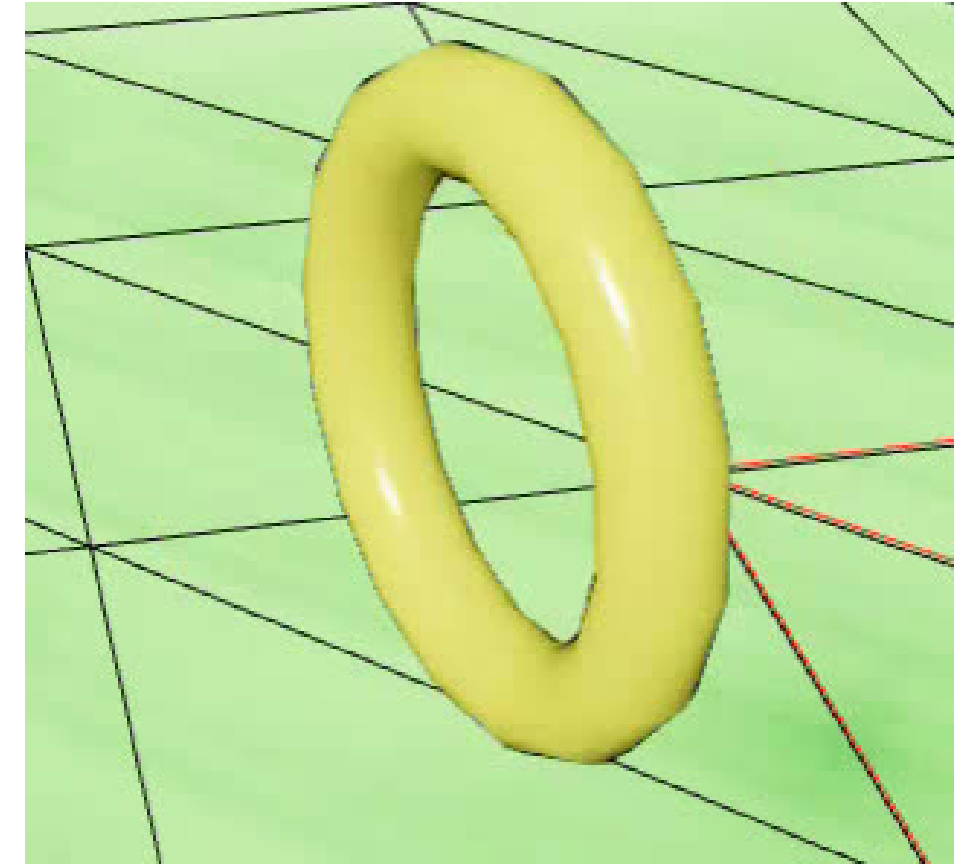
- The information of the status of the game is represented with a dictionary that have rows * columns key and at each key is associated a value
- Legend:
 - **0** empty cell;
 - **1** special object cell;
 - **2** tris cell (empty);
 - **3** tris cell with **X**;
 - **4** tris cell with **O**;

High level logic recap



SPECIAL OBJECTS

- Torus geometry for the shape;
- Check collision between main character and this objects;
- I check every update if index of any special objects in the grid is equal to the index of the main character;
- When a collision is detected the object is removed and the effect activated.



Sonic reference!

CHECK COLLISION

- Both special object and main character have an attribute:
`internal_index` // the index of the cell occupied
- To obtain the coordinate from the index cell I have done the following reasoning:



X (columns)		0	1	2	3	4	5	6	7	8	9	10
Z r o w s	11	12	13	14	15	16	...					

Example:

Given `14`

`coordinate_x = 14 % grid_size`

`coordinate_z = floor(14 / grid_size)`



`coordinate_x = 3`

`coordinate_z = 1`

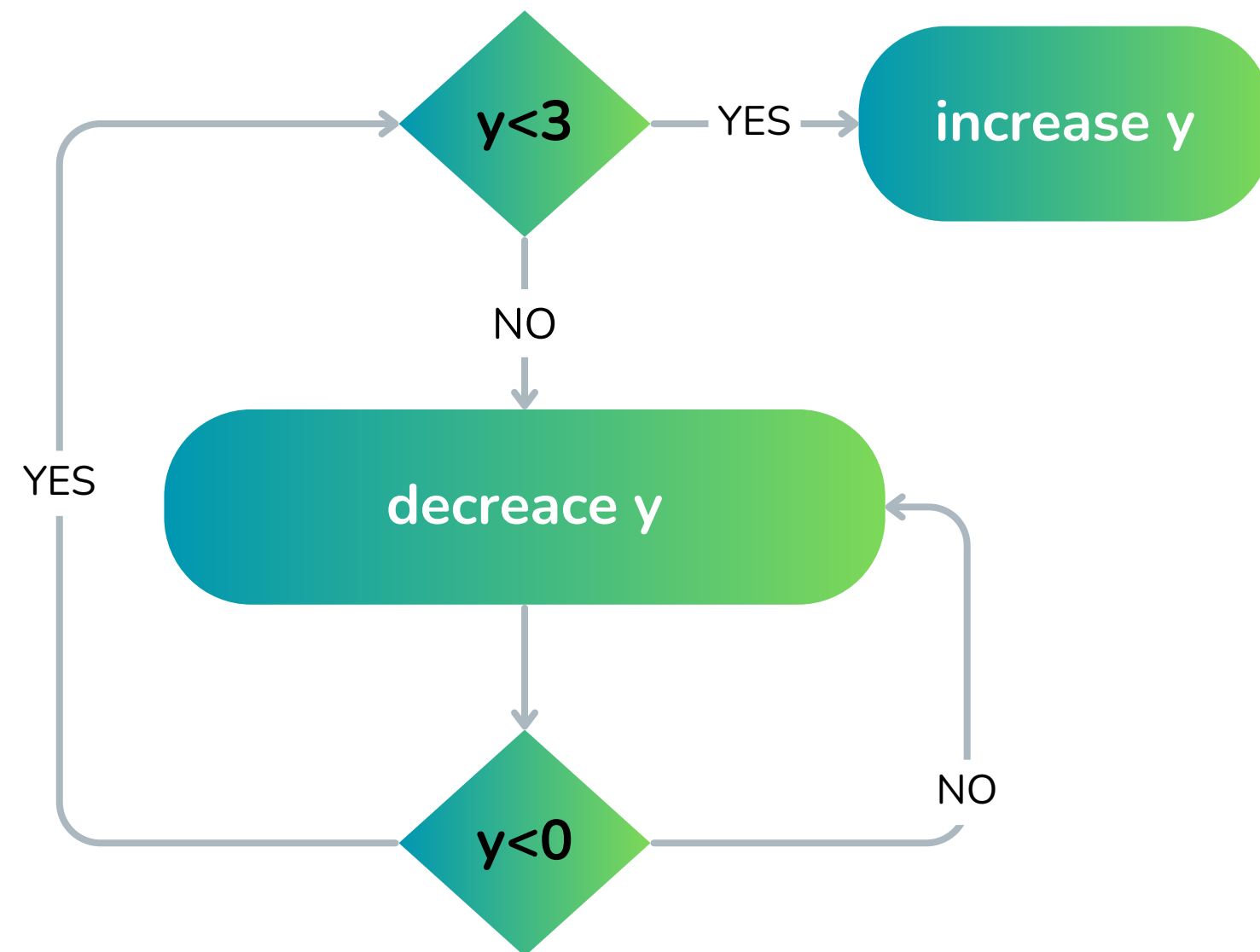
MAIN CHARACTER

- more than 20 methods to manage it!
- loaded with GLTF loader ();
- Very light model, i need to set scale (40,40,40) to see the character well!
- The animations are done with TWIN a tree.js library
- To control it i check every time key event to update where its direction.

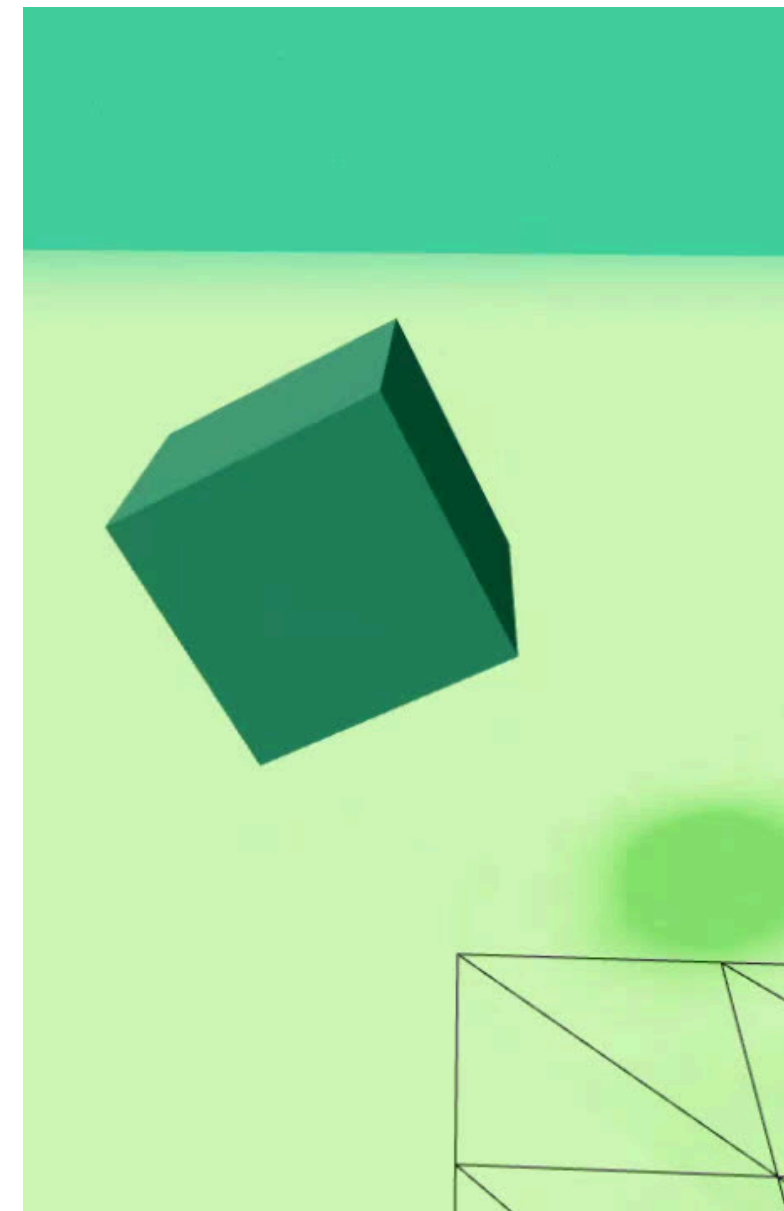


ANIMATIONS - THE CUBIC

LOGIC



RESULT



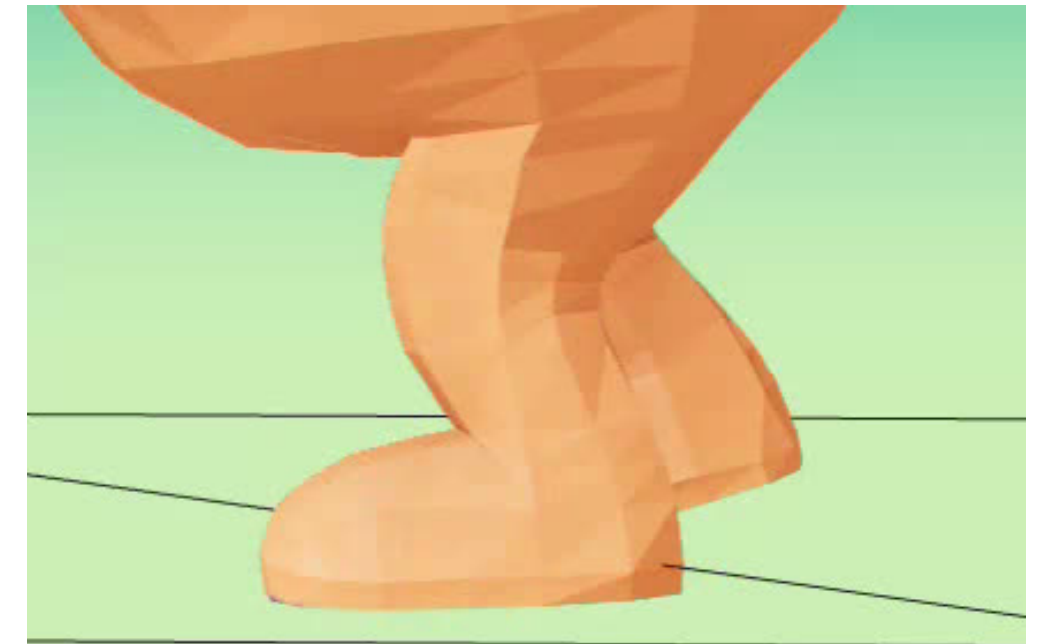
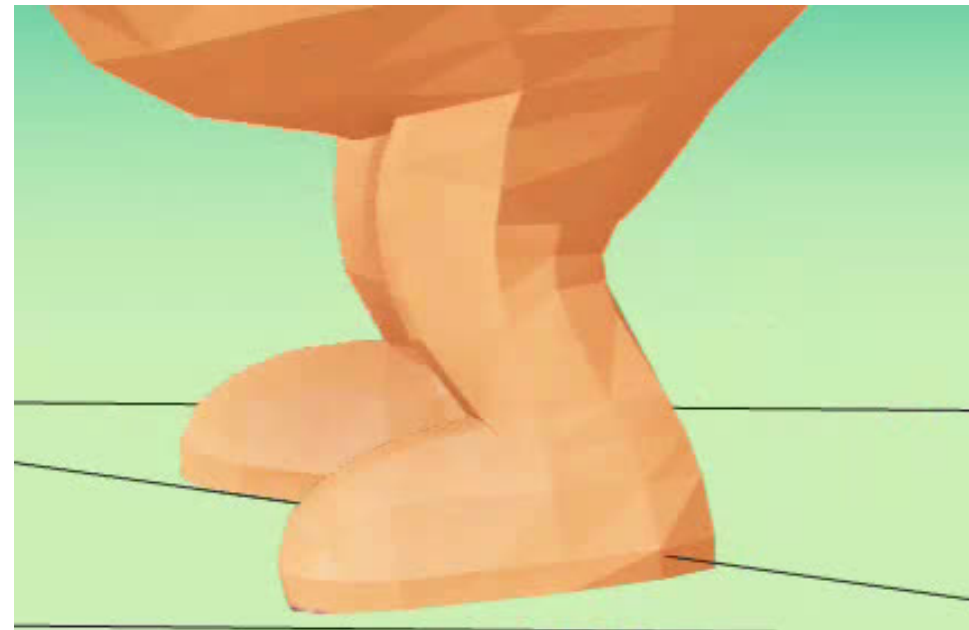
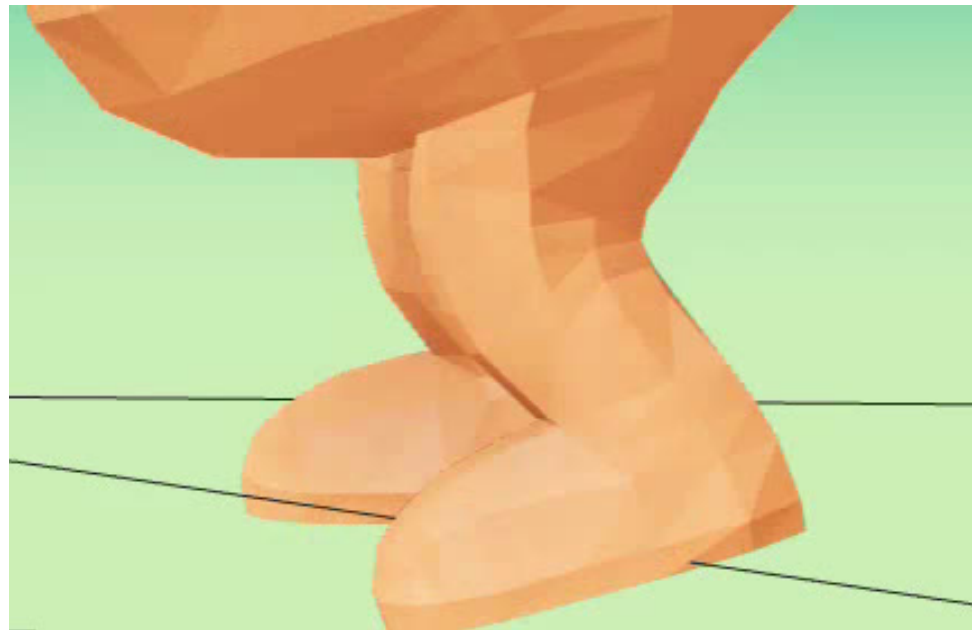
ANIMATIONS - THE NIGHTMARE ONE

- I want create a walk animation of my character.
 - **findBoneByName():**
 - **setArm():** this function rotate right arm and left arm around x-axis to get the character in a straight position



ANIMATIONS - THE NIGHTMARE ONE

- The second step consists in animation of legs with **animateLegs()**, this function is a complex function divided in 3 animations:



ANIMATIONS - THE NIGHTMARE ONE

- The final step to do walk animation is another complex animation in 3 dong by the function **animatedArms()**



ANIMATIONS - THE NIGHTMARE ONE (FINAL RESULT)

